



Model-based ordinations

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Aims

1. **Understand the motivation** for a model-based ordination
2. Be able to **build a simple model based Ordination** in R.
3. Compare model-based results with traditional ordinations.
4. Know **resources** for future analyses of community data



Species associations

- Species may co-occur due to biotic interactions
- Due to similar environmental preferences
- Or because they have a similar history

Either way, it results in correlations between responses

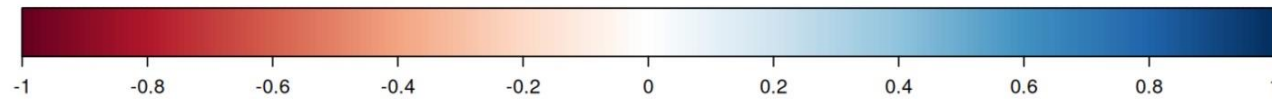


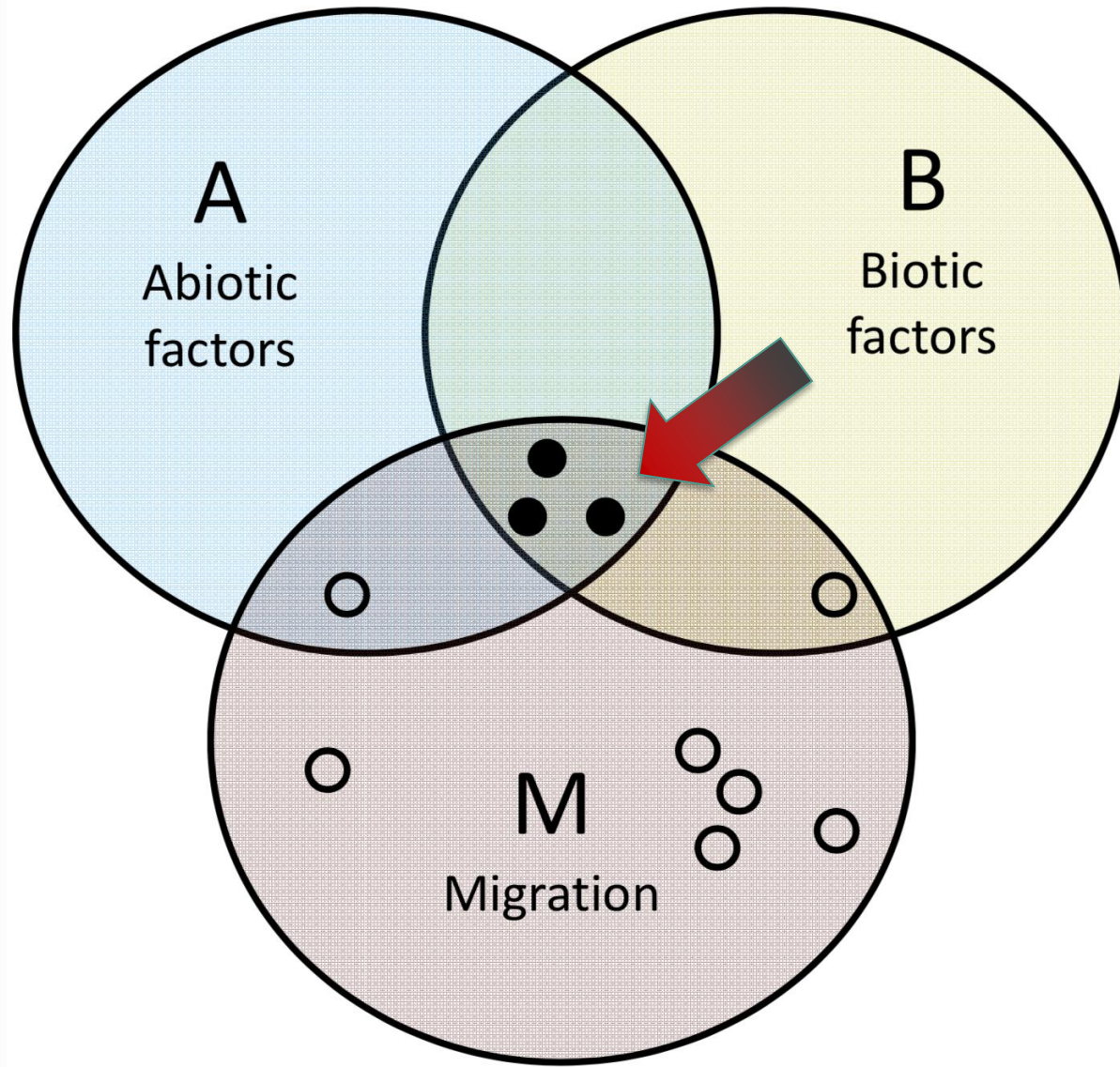


Species associations

Blue: species that are predicted to co-occur

Red: species that are predicted to „avoid“ each other





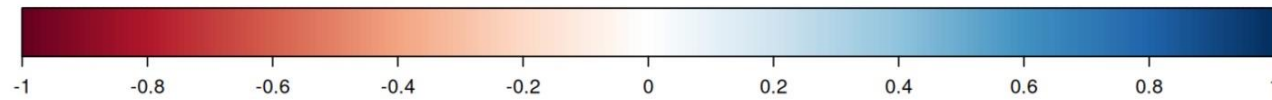
- A – area with suitable climatic conditions
Assuring positive populations growth
- B – area where biotic interactions allow the species to persist, e.g. it is not competitively excluded or are missing mutualistic interactions
- M – area to which the species can migrate
- Population with positive population growth rate
 $r = b - d \geq 0$
- Population with negative population growth rate



Species associations

Blue: species that are predicted to co-occur

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Species associations

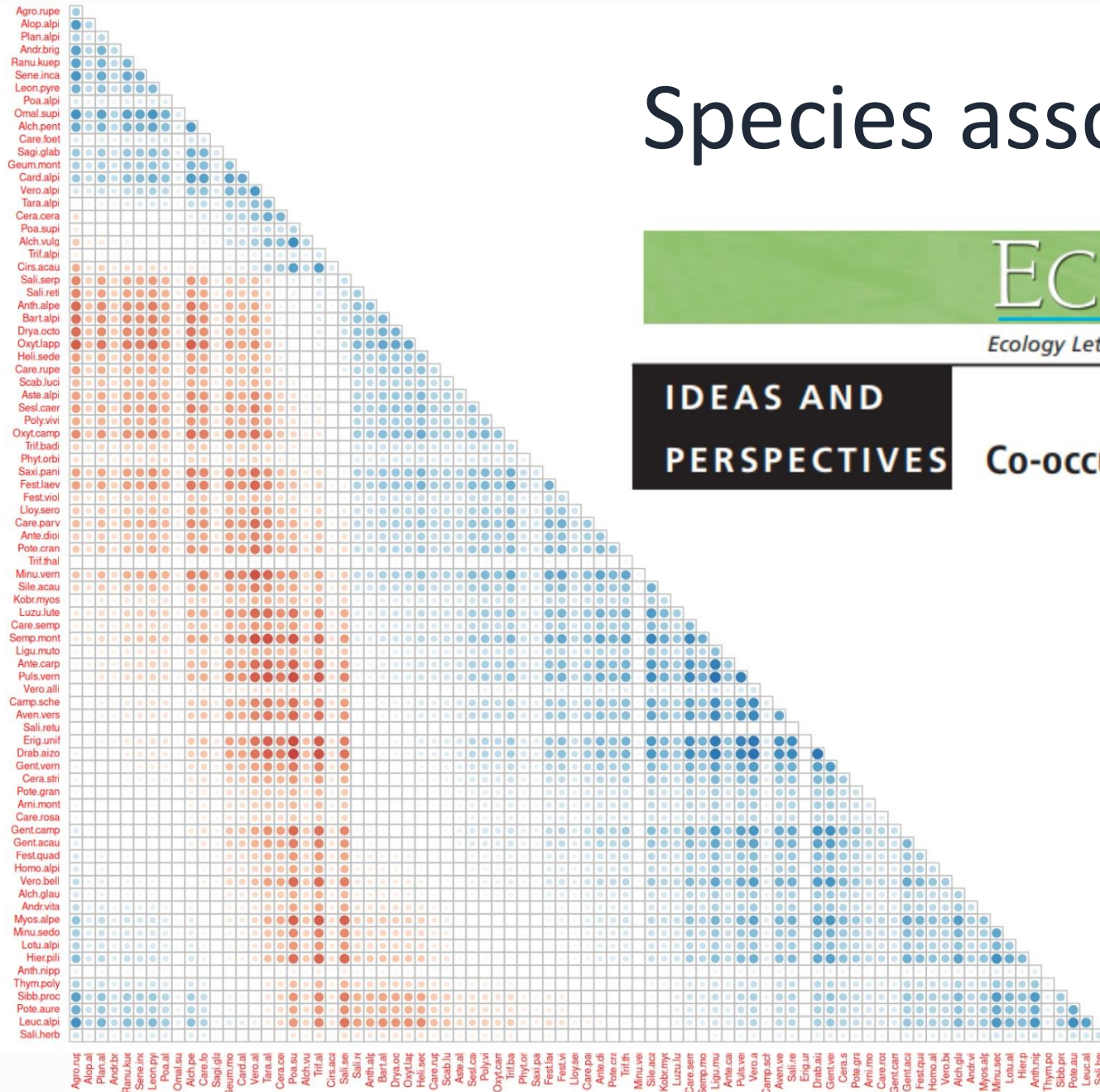
ECOLOGY LETTERS

Ecology Letters, (2020) 23: 1050–1063

doi: 10.1111/ele.13525

IDEAS AND PERSPECTIVES

Co-occurrence is not evidence of ecological interactions





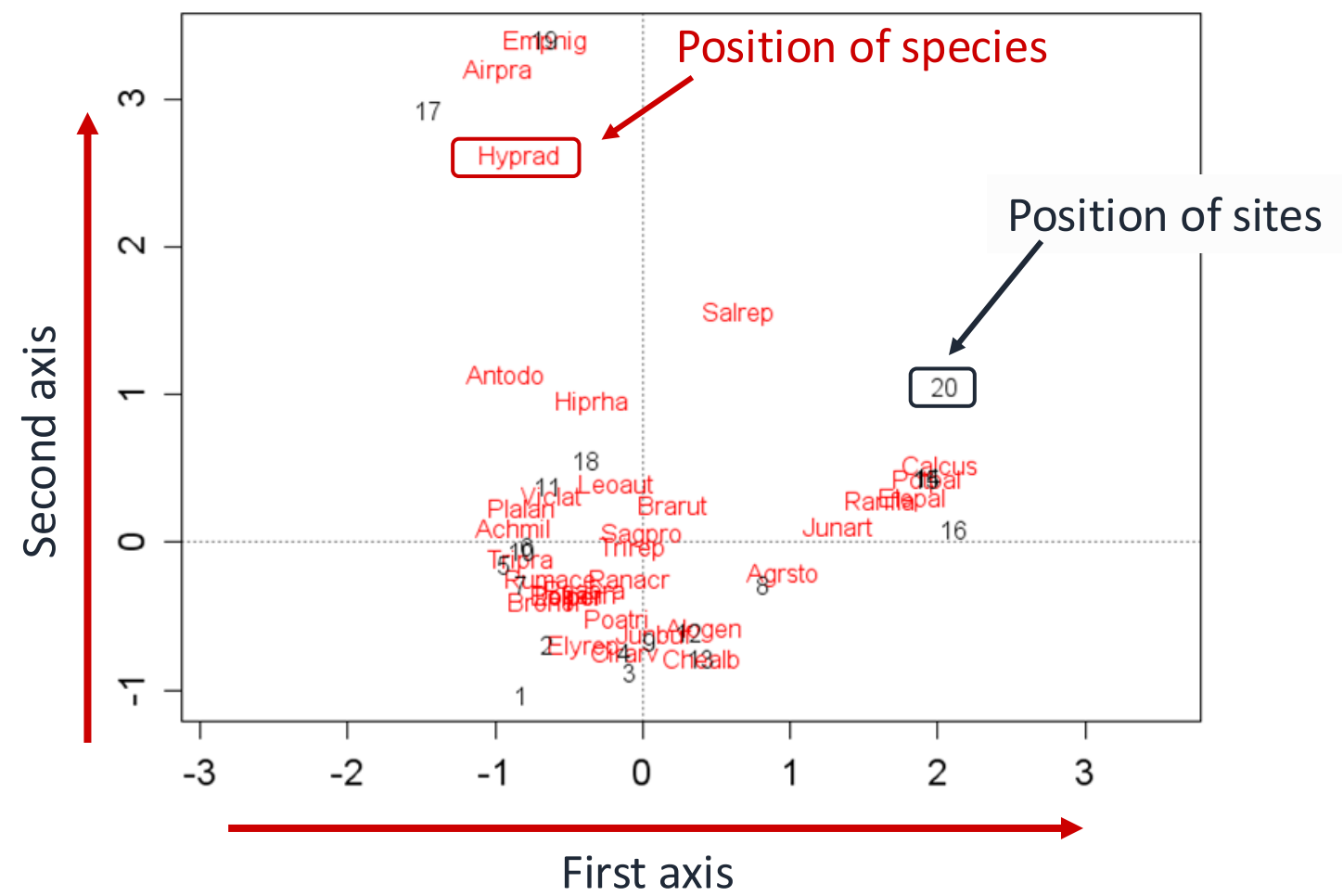
Species associations

- ▶ Difficult to estimate: there are usually too many parameters
- ▶ The number of pairwise associations grows quadratically
 - ▶ 1 with 2 species, 6 for 4 species, 45 for 10 species, 4950 for 100

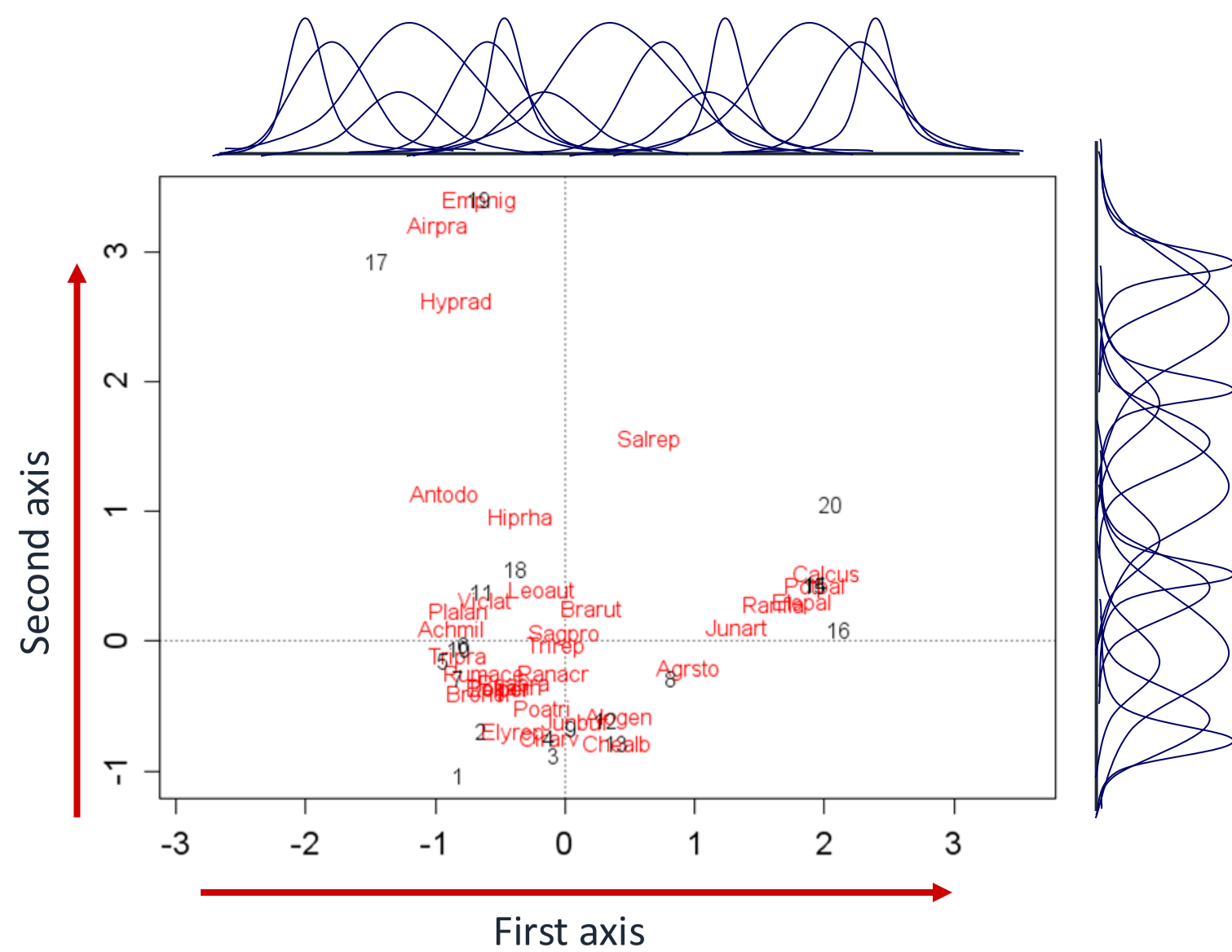
$$\Sigma = \begin{bmatrix} 1 & sp_{12} & \cdots & sp_{1j} \\ sp_{21} & 1 & \cdots & sp_{2j} \\ \vdots & & \ddots & \vdots \\ sp_{j1} & sp_{j2} & \cdots & 1 \end{bmatrix} \quad (4)$$

This very quickly becomes an issue for fitting models

Ordination: Correspondence Analysis



Ordination: Correspondence Analysis



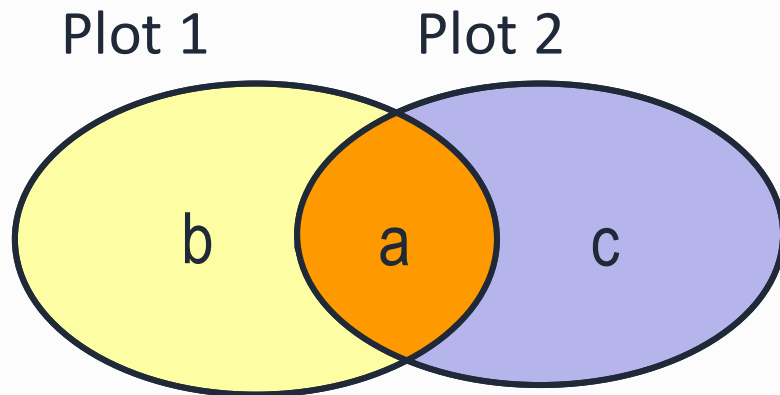
Correspondence Analysis



Correspondence Analysis implicitly uses Chi-square distance/similarity. Which can be expressed as:

$$sim = \frac{(a+b+c+d)(ad-bc)^2}{(a+b)(a+c)(b+d)(c+d)}$$

Yule & Kendall (1950) in Jurasinski (2012)



a: Shared species

b: Plot 1 exclusive species

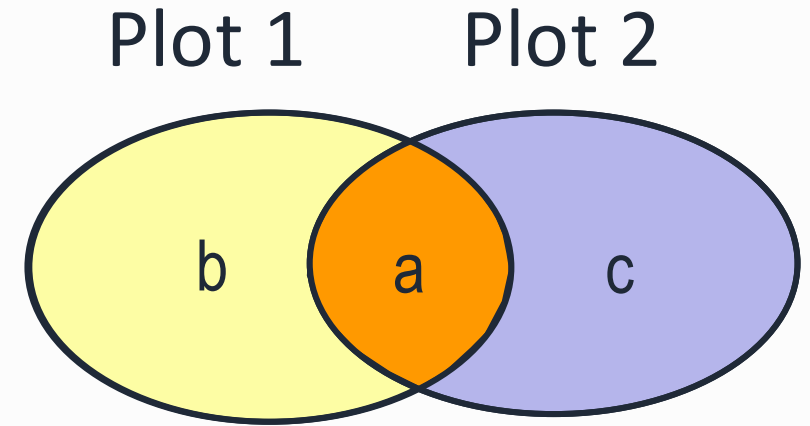
c: Plot 2 exclusive species

d: Species not in the two plots



Similarity- and dissimilarity indices

		<i>Plot 1</i>	
		+	-
<i>Plot 2</i>	+	a	c
	-	b	d



a: Shared species

b: Plot 1 exclusive species

c: Plot 2 exclusive species

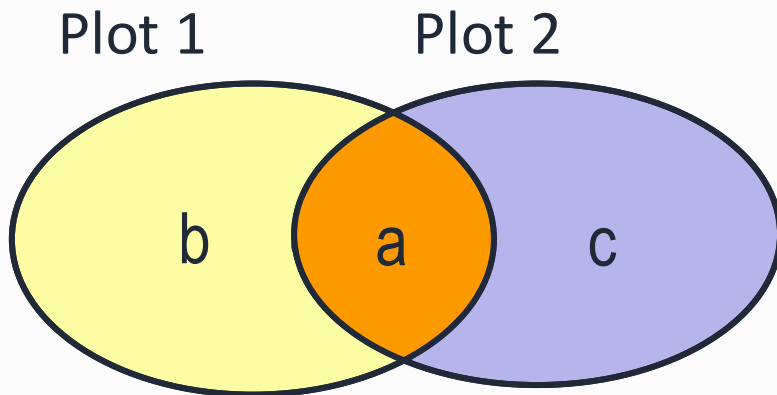
e.g.: Sørensen similarity = $2a / (2a+b+c)$

Correspondence Analysis



Correspondence Analysis implicitly uses Chi-square distance/similarity. Which can be expressed as:

$$\beta_{\text{Sørensen}} = \frac{2a}{2a + b + c}$$

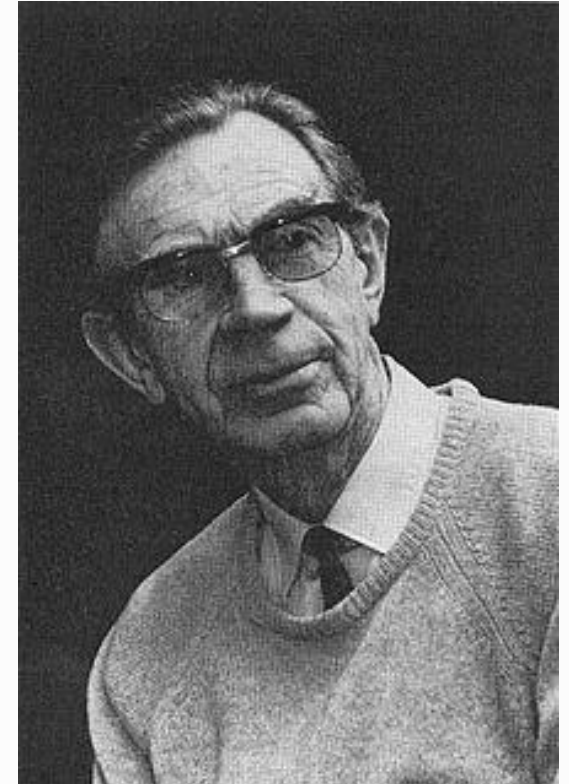


a: Shared species

b: Plot 1 exclusive species

c: Plot 2 exclusive species

d: Species not in the two plots



Thorvald Sørensen

Diversity of indices



$$\beta_{\text{Sørensen}} = \frac{2a}{2a + b + c}$$

$$\beta_{\text{Williams}} = \frac{\min(b, c)}{a + b + c}$$

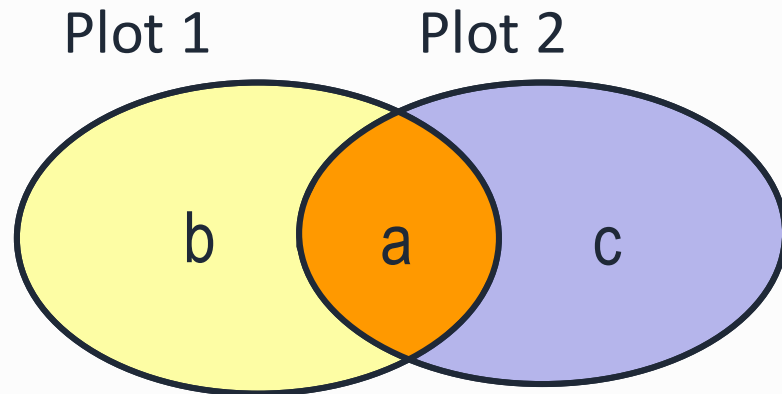
$$\beta_{\text{Jaccard}} = \frac{a}{a + b + c}$$

$$\beta_{\text{Lande}} = \frac{b + c}{2}$$

$$\beta_{\text{Wilson \& Shmida}} = \frac{b + c}{2a + b + c}$$



Diversity of indices



soerensen

$$sim = \frac{2a}{2a+b+c}$$

jaccard

$$sim = \frac{a}{a+b+c}$$

ochiai

$$sim = \frac{a}{\sqrt{(a+b)(a+c)}}$$

mountford

$$sim = \frac{2a}{(a(b+c)+2bc)}$$

whittaker

$$sim = \frac{a+b+c}{\frac{2a+b+c}{2}} - 1$$

lande

$$sim = \frac{b+c}{2}$$

wilsonshmid

$$sim = \frac{b+c}{2a+b+c}$$

cocogaston

$$sim = \frac{b+c}{a+b+c}$$

magurran

$$sim = (2a+b+c)(1 - \frac{a}{a+b+c})$$

harrison

$$sim = \frac{\min(b,c)}{\max(b,c)+a}$$

cody

$$sim = 1 - \frac{a(2a+b+c)}{2(a+b)(a+c)}$$

williams

$$sim = \frac{\min(b,c)}{a+b+c}$$

williams2

$$\frac{(bc)+1}{(a+b+c)^2 - (a+b+c)}$$

harte

$$1 - \frac{2a}{2a+b+c}$$

simpson

$$\frac{\min(b,c)}{\min(b,c)+a}$$

lennon

$$\frac{2|b-c|}{2a+b+c}$$

weiher

$$sim = b + c$$

ruggiero

$$sim = \frac{a}{a+c}$$

lennon2

$$sim = 1 - \left[\frac{\log\left(\frac{2a+b+c}{a+b+c}\right)}{\log 2} \right]$$

routledge

$$sim = \frac{(a+b+c)^2}{(a+b+c)^2 - 2bc}$$

routledge

toolong, see below

routledge

$$sim = e^{routledge} - 1$$

sokal1

$$sim = \frac{a}{a+2(b+c)}$$

dice

$$sim = \frac{a}{\min((b+a), (c+a))}$$

kulczynsky

$$sim = \frac{a}{b+c}$$

Soerensen (1948)

Jaccard (1912)

Ochiai (1957), Shi (1993)

Mountford (1962), Shi (1993)

Whittaker (1960), Magurran (1988)

Lande (1996)

Wilson & Shmida (1984)

Colwell & Coddington (1948), Gaston et al. (2001)

Magurran (1988)

Harrison et al. (1992), Koleff et al. (2003)

Cody (1993)

Williams (1996), Koleff et al. (2003)

Williams (1996), Koleff et al. (2003)

Harte & Kinzig (1997), Koleff et al. (2003)

Simpson (1949), Koleff et al. (2003)

Lennon et al. (2001), Koleff et al. (2003)

Weiher & Boylen (1994)

Ruggiero et al. (1998), Koleff et al. (2003)

Lennon et al. (2001), Koleff et al. (2003)

Routledge (1977), Magurran (1988)

Routledge (1977), Wilson & Shmida (1984)

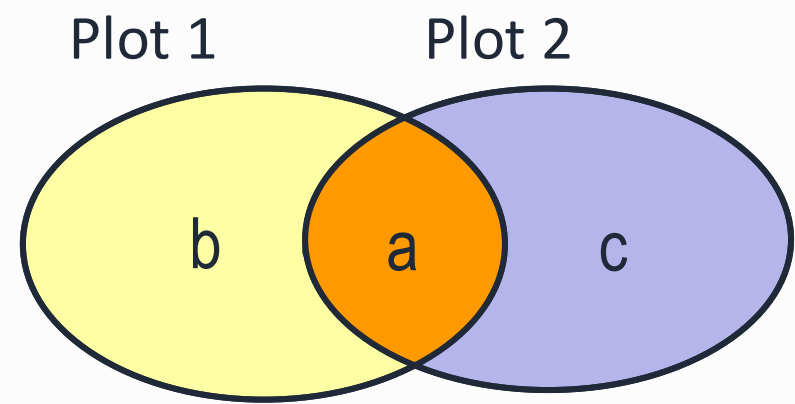
Routledge (1977)

Sokal & Sneath (1963)

Association index of Dice (1945), Wolda (1981)

Oosting (1956), Southwood (1978)

Diversity of indices



kulcz2insky	$sim = \frac{\frac{a}{2}(2a+b+c)}{(a+b)(a+c)}$
mcconnagh	$sim = \frac{\frac{a^2-bc}{a^2}}{(a+b)(a+c)}$
simpson2	$sim = \frac{\frac{a}{a+b}}{\frac{a}{a+b} + \frac{a}{a+c}}$
legendre2	$sim = \frac{\frac{3a}{3a+b+c}}{\frac{a}{a+b} + \frac{a}{a+c}}$
fager	$sim = \frac{\frac{a}{\sqrt{n_1 n_2}}}{2 * \sqrt{n_2}}$
maarel	$sim = \frac{\frac{2a-(b+c)}{2a+b+c}}{\frac{a}{a+b} + \frac{a}{a+c}}$
lamont	$sim = \frac{\frac{a}{2a+b+c}}{\frac{a}{a+b} + \frac{a}{a+c}}$
johnson	$sim = \frac{\frac{a}{2b}}{\frac{a}{a+b} + \frac{a}{a+c}}$
sorgenfrei	$sim = \frac{\frac{a^2}{(a+b)(a+c)}}{\frac{a}{a+b} + \frac{a}{a+c}}$
johnson2	$sim = \frac{\frac{a}{a+b} + \frac{a}{a+c}}{\frac{a}{a+b} + \frac{a}{a+c}}$

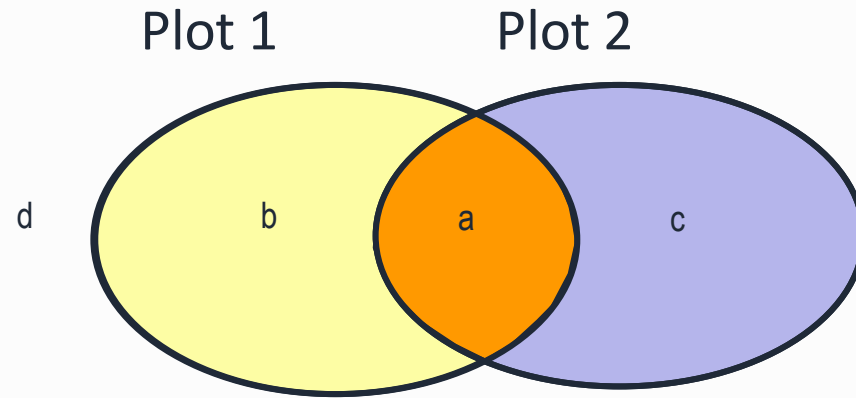
Oosting (1956), Southwood (1978)
 Hubalek (1982)
 Simpson (1960), Shi (1993)
 Legendre & Legendre (1998)
 Fager (1957), Shi (1993)
 van der Maarel (1969)
 Lamont and Grant (1979)
 Johnson (1971)
 Sorgenfrei (1959)
 Johnson (1967)

Computable symmetric indices (including unshared species):

manhattan	$sim = \frac{\frac{b+c}{a+b+c+d}}{\frac{a+d}{a+b+c+d}}$
simplematching	$sim = \frac{\frac{a+b+c+d}{a+b+c+d}}{\frac{a+b+c+d}{a+b+c+d}}$
margaleff	$sim = \frac{\frac{a(a+b+c+d)}{(a+b)(a+c)}}{\frac{ad-bc}{ad-bc}}$
pearson	$sim = \frac{\frac{a+d}{\sqrt{(a+b)(a+c)(d+b)(d+c)}}}{\frac{ad-bc}{ad-bc}}$
roger	$sim = \frac{\frac{a+d}{a+2(b+c)+d}}{\frac{ad-bc}{ad-bc}}$
baroni	$sim = \frac{\frac{\sqrt{ad+c}}{\sqrt{ad+a+b+c}}}{\frac{ad-bc}{ad-bc}}$
dennis	$sim = \frac{\frac{\sqrt{(a+b+c+d)(a+b)(a+c)}}{(a+b+c+d)(-\frac{a}{2})^2}}{\frac{ad-bc}{ad-bc}}$
fossum	$sim = \frac{\frac{(a+b)(a+c)}{a-(b+c)+d}}{\frac{ad-bc}{ad-bc}}$
gower	$sim = \frac{\frac{a}{a+b+c+d}}{\frac{ad-bc}{ad-bc}}$
legendre	$sim = \frac{\frac{a}{a+b+c+d}}{\frac{ad-bc}{ad-bc}}$
sokal2	$sim = \frac{\frac{ad}{\sqrt{(a+b)(a+c)(d+b)(d+c)}}}{\frac{ad-bc}{ad-bc}}$
sokal3	$sim = \frac{\frac{2a+2d}{(a+d)+(a+b+c+d)}}{\frac{ad-bc}{ad-bc}}$
sokal4	$sim = \frac{\frac{\frac{a+d}{b+c}}{(a+b+c+d)(ad-bc - \frac{a+b+c+d}{2})^2}}{\frac{ad-bc}{ad-bc}}$
stiles	$sim = \frac{\frac{ad-bc}{ad+bc}}{\frac{ad-bc}{ad-bc}}$
yule	$sim = \frac{\frac{4(ad-bc)}{(a+b)^2+(b+c)^2}}{\frac{ad-bc}{ad-bc}}$
michael	$sim = \frac{\frac{(a+d)-(b+c)}{(a+d)-(b+c)}}{\frac{ad-bc}{ad-bc}}$
hamann	$sim = \frac{\frac{N}{(aN-2n_2)}}{\frac{ad-bc}{ad-bc}}$
forbes	$sim = \frac{\frac{2(ad-bc)^2}{(a+b)(a+c)(b+d)(c+d)}}{\frac{ad-bc}{ad-bc}}$
chisquare	$sim = \frac{\frac{(ad-bc)}{(a+c)(b+d)}}{\frac{ad-bc}{ad-bc}}$
peirce	$sim = \frac{\frac{a-(a+b)(a+c)}{(a+b)(a+c)(b+d)(c+d)}}{\frac{ad-bc}{ad-bc}}$
eyraud	$sim = \frac{\frac{a-(a+b)(a+c)}{(a+b)(a+c)(b+d)(c+d)}}{\frac{ad-bc}{ad-bc}}$

Mean Manhattan, Legendre & Legendre (1998)
 Sokal & Michener 1958
 Clifford & Stevenson (1975)
 Phi of Pearson, Gower & Legendre (1986)
 Rogers & Tanimoto (1960), Gower & Legendre (1986)
 Baroni-Urbani & Buser (1976), Wolda (1981)
 Holliday et al. (2002)
 Holliday et al. (2002)
 Gower & Legendre (1986)
 Gower & Legendre (1986)
 Sokal & Sneath (1963)
 Sokal & Sneath (1963)
 Sokal & Sneath (1963)
 Stiles (1946)
 Yule & Kendall (1973)
 Michael (1920), Shi (1993)
 Hamann (1961)
 Forbes (1925), Shi (1993)
 Yule & Kendall (1950)
 Peirce (1984) in Shi (1993)
 Eyraud (1936) in Shi (1993)

Similarity- and dissimilarity indices



a: Shared species

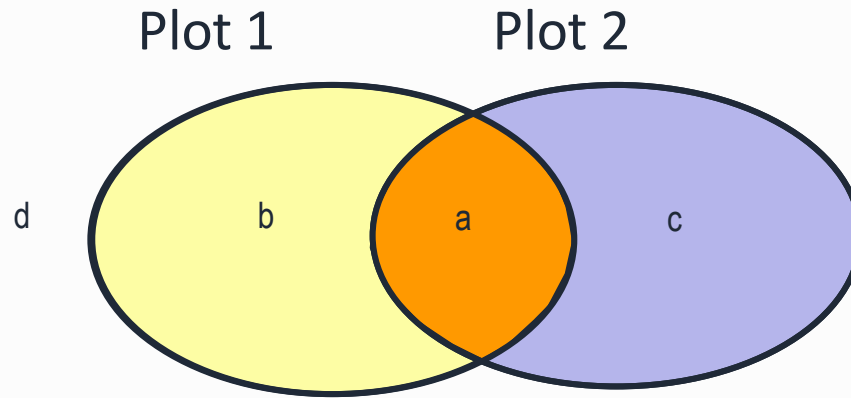
b: Plot 1 exclusive species

c: Plot 2 exclusive species

d: Further species



Similarity- and dissimilarity indices



e.g.: Sørensen similarity = $2a / (2a+b+c)$

	V_1	V_2	V_3	V_4
O_1	$V_{1,1}$	$V_{1,2}$	$V_{1,3}$	$V_{1,4}$
O_2	$V_{2,1}$	$V_{2,2}$	$V_{2,3}$	$V_{2,4}$
O_3	$V_{3,1}$	$V_{3,2}$	$V_{3,3}$	$V_{3,4}$
O_4	$V_{4,1}$	$V_{4,2}$	$V_{4,3}$	$V_{4,4}$
O_5	$V_{5,1}$	$V_{5,2}$	$V_{5,3}$	$V_{5,4}$
O_6	$V_{6,1}$	$V_{6,2}$	$V_{6,3}$	$V_{6,4}$

Raw data



similarity
matrix

	O_1	O_2	O_3	O_4	O_5	O_6
O_1	1					
O_2	$s_{2,1}$	1				
O_3	$s_{3,1}$	$s_{3,2}$	1			
O_4	$s_{4,1}$	$s_{4,2}$	$s_{4,3}$	1		
O_5	$s_{5,1}$	$s_{5,2}$	$s_{5,3}$	$s_{5,4}$	1	
O_6	$s_{6,1}$	$s_{6,2}$	$s_{6,3}$	$s_{6,4}$	$s_{6,5}$	1

Sites

Community data

Species

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	
Achimill		4	4			1																			1					3				
Agrocani							4	5									6			1														
Agrocapi	3	4	2													8				1	1						4		5	3				
Agrorepe		1	1		7	5			4		4					9	3			3	5					5			9	5				
Agrostol						6			4		12									5		1				2	3					8		
Alopgeni									4		1									2					3		5				6			
Alopprat		1			7	7	1		2		5		1	1	4	2			4	5		3	2		5	1	1	3	2					
Anthodor	6					5							1				1				1						4			4				
Caltpalu				1													1	5					6											
Cardprat			2			2	1					1	3						1	1	2	1							1					
Caredist																		1		2														
Caregrac			4			5	8			1		7	2				4	3				1	2			2							1	
Careprae		2	2		1	1										2			1						2		1	3	1					
Carevesi			4			1	4										1	7			2													
Carevulp							1		1		1	1	1						3		1				1									
Cirsarve							1						4							3										1				
Cirslanc																							1											
Cniddubi																													2					
Desccesp													8				1		5	4									3	1				
Eleounig											1	2								1	1													
Eropvern	1					1																					1	1						
Euphesul			1																									1		1				
Festprat						2	1																						1					
Galipalu				1			2	1		1	1	3	2				3		1		1	3	1	1	2						1			
GaliverA	1	1	3										4														4		1	4				
Glycflui							1												2					7							2			
Glycmaxi							1				2												1	2	3		1				1	11	10	
Holclana						4	2						4				1		2	6										1				
Junceffu			1			1	1						1							4	1					1								
Lathprat												1	1								1									1				
Lotucorn		1	1			1																1								1				
Lychflos			1			2															1													
Phalarun			5			1	2	2	1	11	8	2					1		1	2	1	1	6			1					1			
Planinte								3	3	1									1							1								
Poa palu				1		1	1		11		4		1		1		1		6	2		4	1		6	3								
Poa praA	3	2		1	5	6	1							3		1			1		4		2		2			4	2					
Poa triv			1		2	1	1	4												2		2	3		3	1								
Poteanse			1			2									1				1		2													
Poterept							1												1							1								
Ranuflam			1			1	3						1								1							3						
Ranurepe			4	1	1	1	2	4	1	4			1	3			3		5	4	4	2	6		9	9								
Roriamph								4		7																1		1						
RorisylA							1	2	1							1			1				3		4									
Rumeacet				1			4															4						1			1			
Rumethyr	1	5	2			3	5									3										2	4			1				
Siumlati				1			2				1							3					2	1			1							
Stelpalu			2			1				1	1	1					1	1	2			1	1											
Sympoffi																							3	1										
TaraoffA			1	1	1	1					1					1			1	1		1			2				1	1				
Trifrepe				2			3												4	4	1	1												
Vicicrac				1			1				9		4																	1	2			
Vicilath			1																															
Vicitetr		1	1										2														1	1						



similarity
matrix

	O ₁	O ₂	O ₃	O ₄	O ₅	O ₆
O ₁	1					
O ₂	s _{2,1}	1				
O ₃	s _{3,1}	s _{3,2}	1			
O ₄	s _{4,1}	s _{4,2}	s _{4,3}	1		
O ₅	s _{5,1}	s _{5,2}	s _{5,3}	s _{5,4}	1	
O ₆	s _{6,1}	s _{6,2}	s _{6,3}	s _{6,4}	s _{6,5}	1



	O_1	O_2	O_3	O_4	O_5	O_6
O_1	0					
O_2	$d_{2,1}$	0				
O_3	$d_{3,1}$	$d_{3,2}$	0			
O_4	$d_{4,1}$	$d_{4,2}$	$d_{4,3}$	0		
O_5	$d_{5,1}$	$d_{5,2}$	$d_{5,3}$	$d_{5,4}$	0	
O_6	$d_{6,1}$	$d_{6,2}$	$d_{6,3}$	$d_{6,4}$	$d_{6,5}$	0

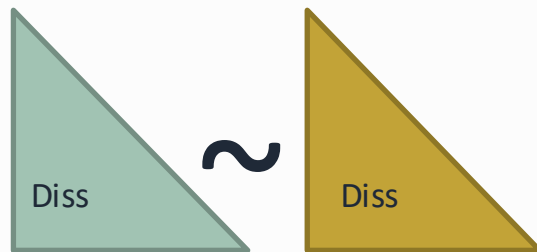
Dissimilarity matrix



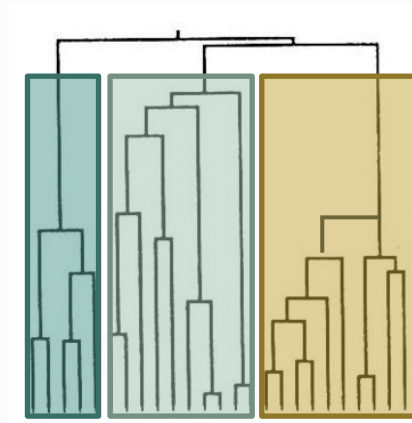
or



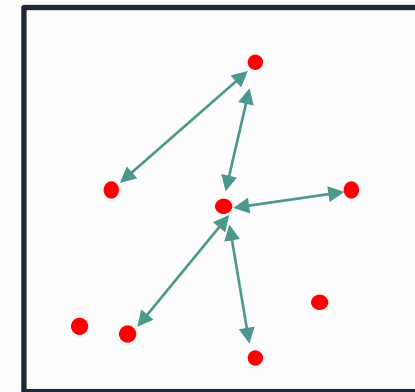
or



Relate to other
dissimilarity matrix

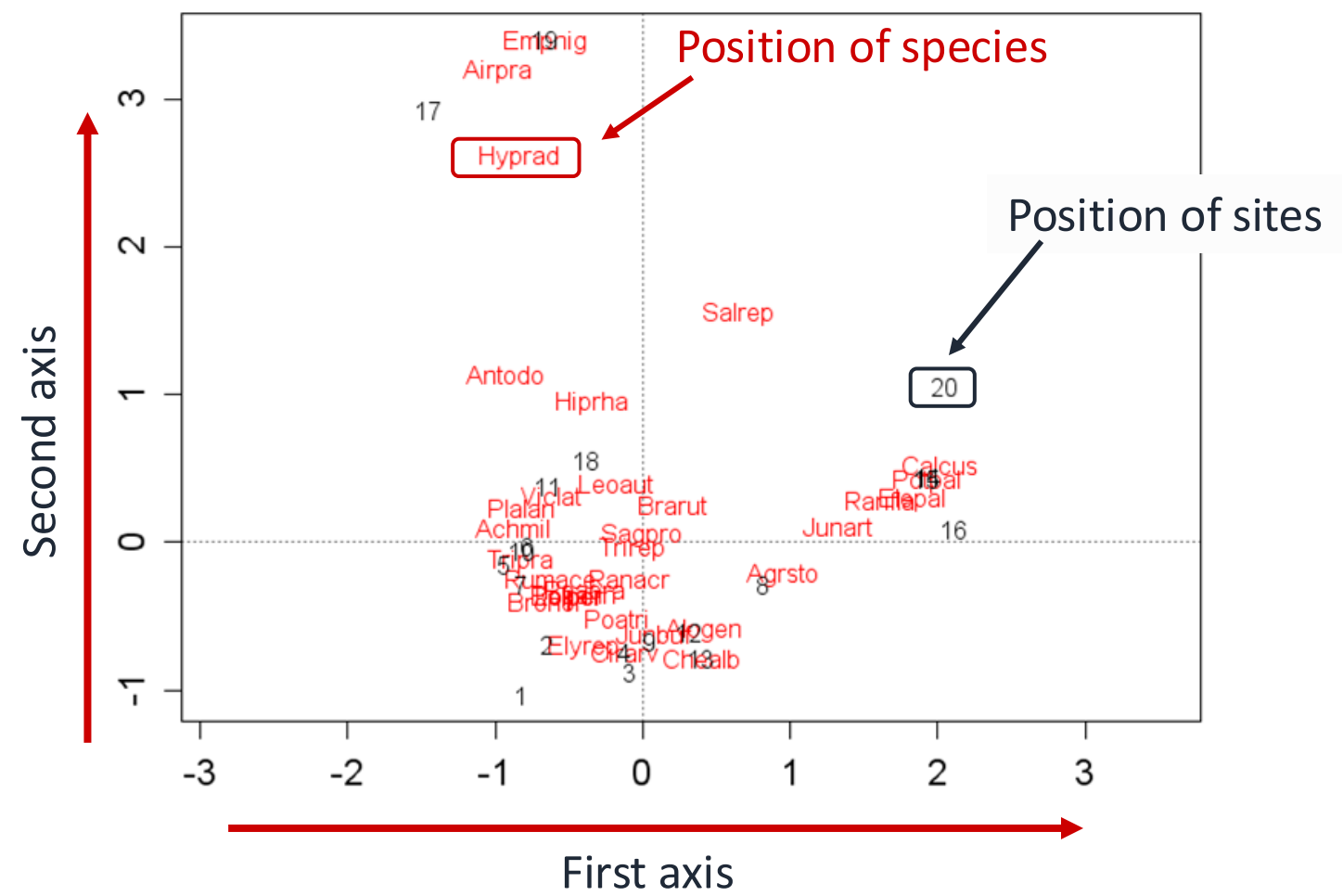


Classification



Dimension reduction
(e.g. NMDS)

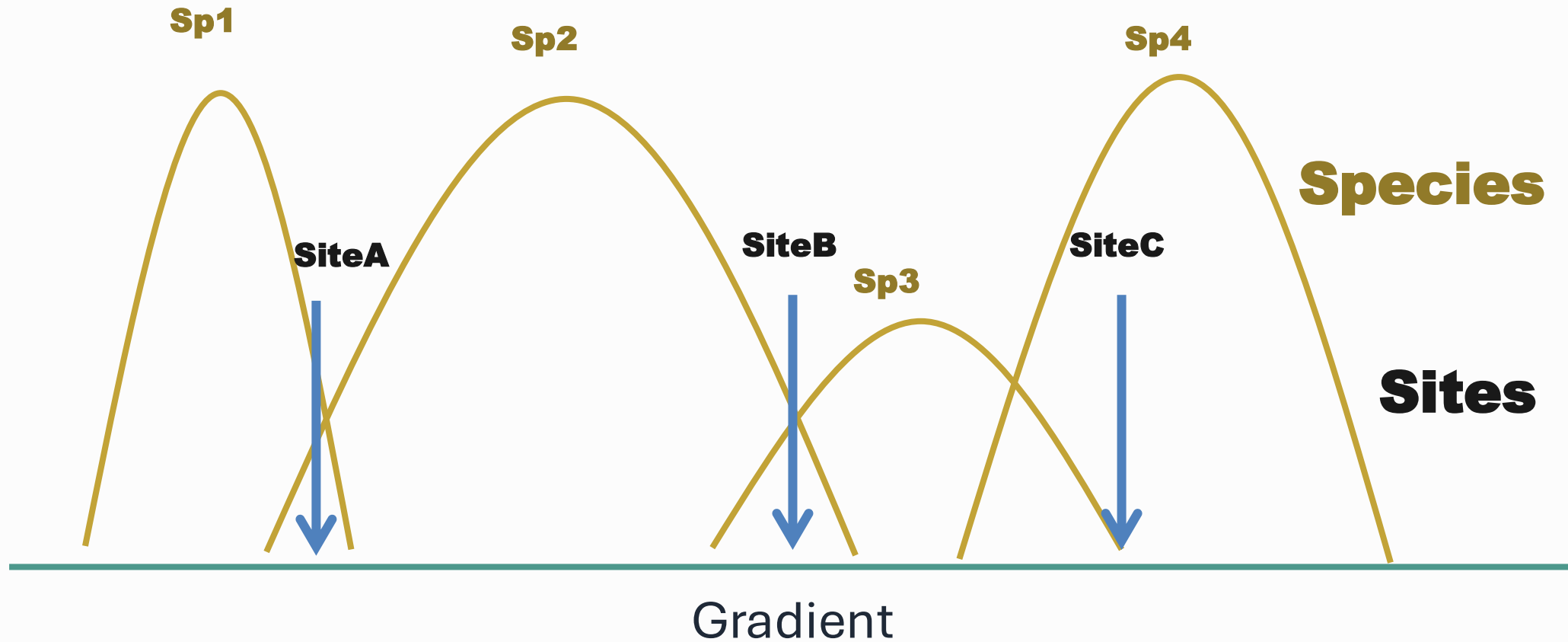
Ordination: Correspondence Analysis





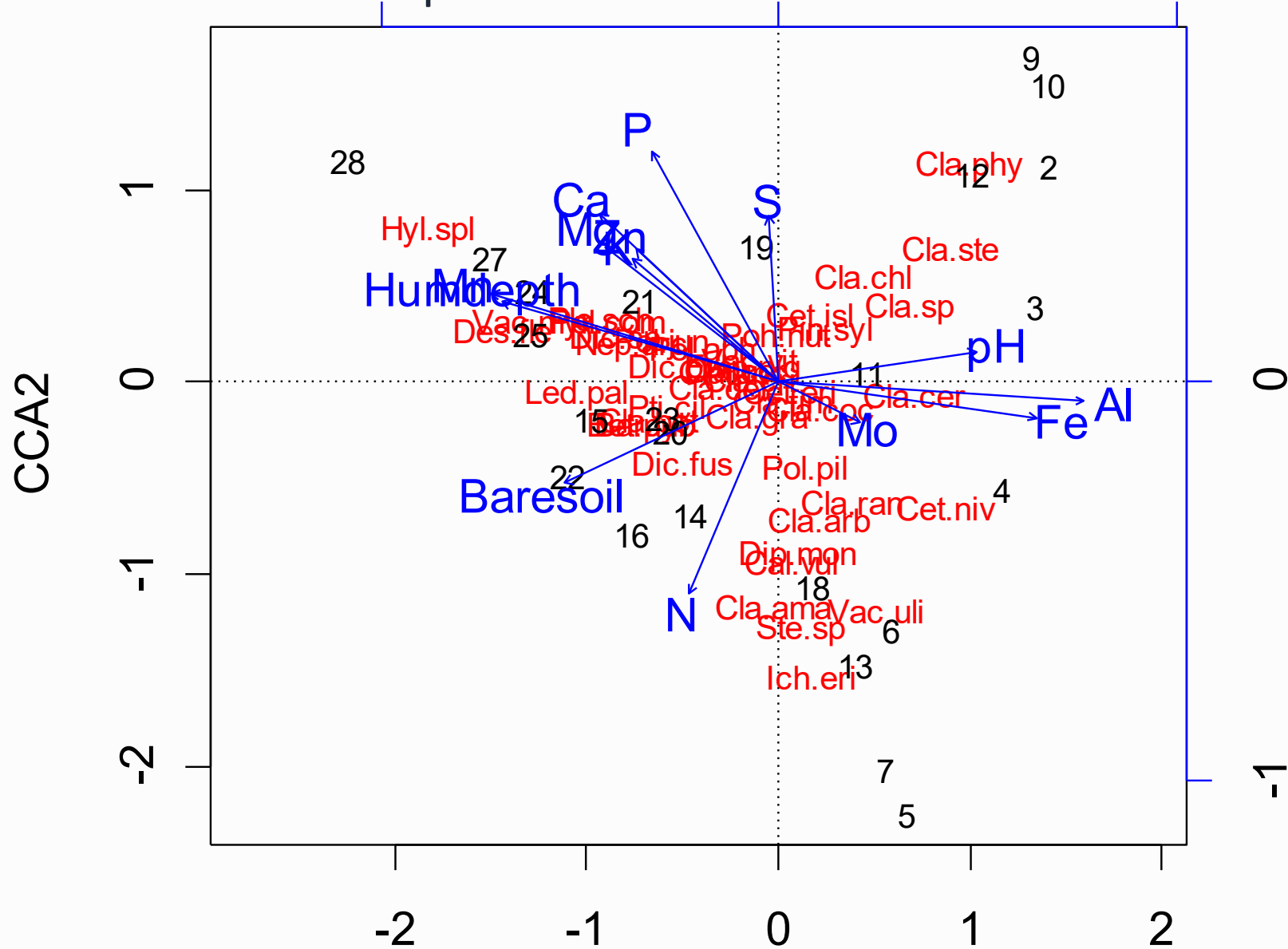
Ordination - Correspondence Analysis

What is it trying to do?



If you knew the species or the sites you could position the other

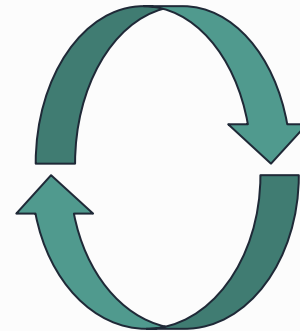
CCA – with environmental predictors



CCA methods: Linking in the Environmental Data

1. Calculate species scores by weighted averaging of site scores
2. Calculate updated site scores by weighted averaging of species scores

(So far, like CA)



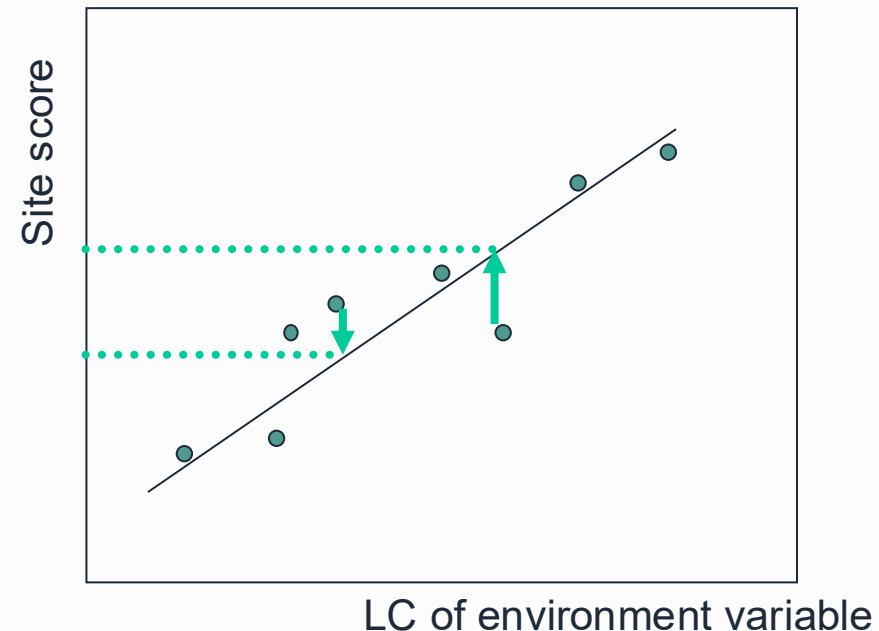
iterations

CCA methods: Linking in the Environmental Data

3. Multiple regression of site scores on environmental variables and calculation of regression coefficients

4. Calculate new site scores as fitted values from this regression

Transforms the ordination axis into a linear combination (LC) of environmental variables

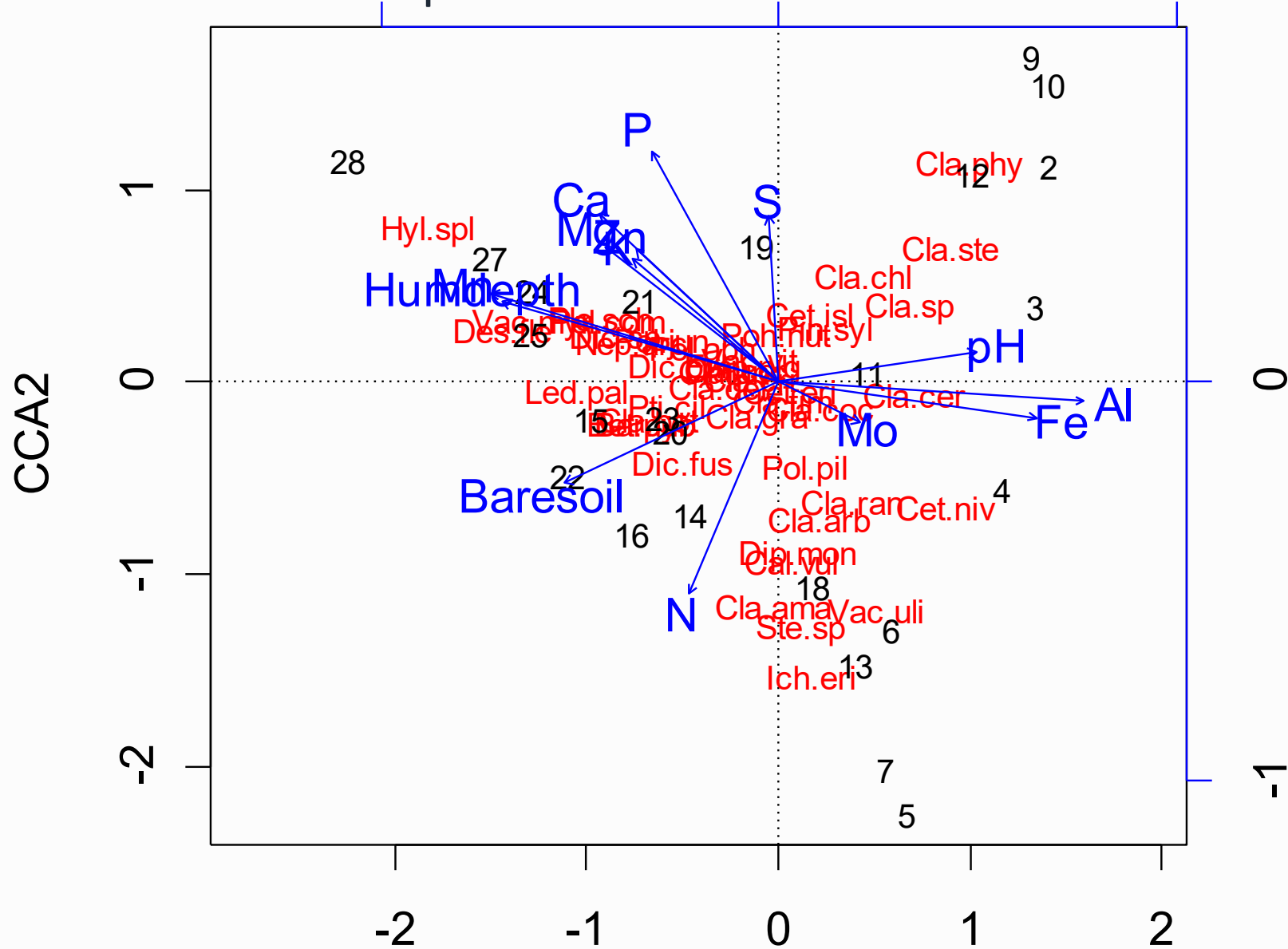


Overview on ordination methods

	Indirect (no environmental data)	Direct (with environmental data)
Linear relation	PCA (PCoA) Principal Components/ Coordinates Analysis	RDA Redundancy Analysis
Unimodel relation	CA (DCA) (Detrended) Correspondence Analysis	CCA Canonical CA
Any relation	NMDS (Non-metric multidimensional scaling)	

cca(varespec, varechem)

CCA – with environmental predictors



No uncertainty

Ordination

- Ordination scores have no standard errors
- You cannot propagate uncertainty in ordination scores to downstream analyses

No prediction

- Distance-based methods do not define a model for the data.
- Cannot forecast community composition under future climate
- Species richness cannot be derived from an NMDS ordination

No random effects (like mixed models)

- Repeated measurements of sites
- Plots nested in sites
- Spatial or temporal autocorrelation

As generalized linear mixed model (GLMM):

```
Model <- glvm(Y, X, formula = ~(bio_1 + bio_12 + slp | 1),  
family = "poisson", num.lv = 0)
```

Y: species abundances
X: environmental variables
num.lv: number of latent variables

As generalized linear mixed model (GLMM):

```
Model <- gllvm(Y, X, formula = ~(bio_1 + bio_12 + slp | 1),  
family = "poisson", num.lv = 0)
```

As Ordination:

```
Model <- gllvm(Y, family = "poisson", num.lv = 2)
```

Y: species abundances
X: environmental variables
num.lv: number of latent variables

As generalized linear mixed model (GLMM):

```
Model <- glvm(Y, X, formula = ~(bio_1 + bio_12 + slp | 1),  
family = "poisson", num.lv = 0)
```

As Ordination:

```
Model <- glvm(Y, family = "poisson", num.lv = 2)
```

Combination:

```
Model <- glvm(Y, X, formula = ~(bio_1 + bio_12),  
family = "poisson", num.lv = 2)
```

Questions?



Tutorials

- Examples and documentation on GLLVM by Jenni Niku et al:
<https://jenniniku.github.io/gllvm/index.html>
- One Toolbox, Many Tools: A Practitioner's Guide to Model-based Ordination for Community Ecology:
https://ecoevorxiv.org/repository/view/11688/?utm_source=chatgpt.com
- Getting started with sjSDM by Maximilian Pichler & Florian Hartig:
https://stat.ethz.ch/CRAN/web/packages/sjSDM/vignettes/sjSDM_Introduction.html?utm_source=chatgpt.com
- Web-page with more traditional resources on Community Data in R by David Zelený: www.davidzeleny.net/anadat-r/doku.php/en:start



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Overview

Repositories 34

Projects

Packages

Stars 58



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 R  63  24



 GLLVM-workshop Public

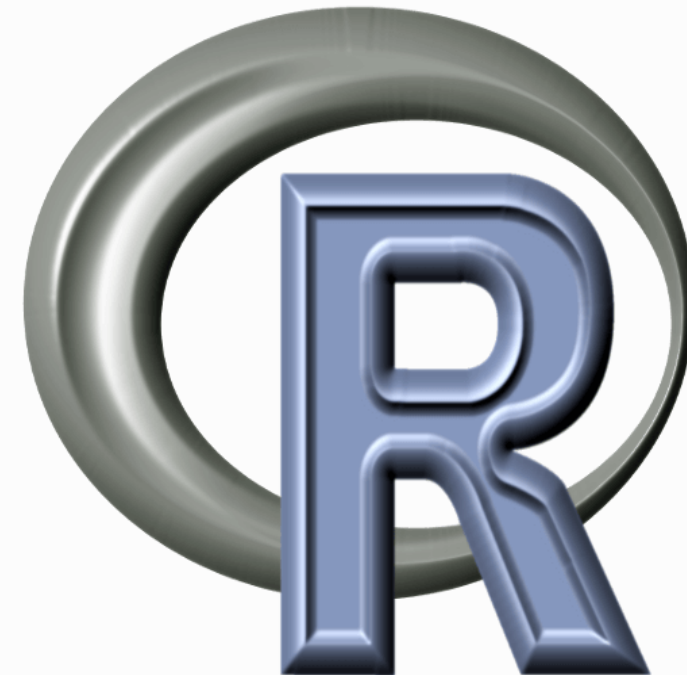
Physalia workshop on Generalized Linear Latent Variable Models by Bert van der Veen

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 Nof2025GLLVMworkshop Public

GLLVM Workshop material for the Norwegian Oikos conference in 2025

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Find more examples and documentation on:
<https://jenniniku.github.io/gllvm/index.html>

Thank you!